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Research Article

The decline in water level and discharge of Lake Toba of North Sumatera, Indonesia, affected by land degradation

Indra Agus Riyanto[1*], Heru Hendrayana[2], Yuli Widyaningsih[3], Galih Dwi Jayanto[4], Azmin Nuha[5], Sabdo Sumartono[6]

[1] Department of Geography Information Science, Faculty of Science, Technology, Engineering, and Mathematics, Universitas Maha Karya Asia, Yogyakarta, Indonesia
[2] Department of Geological Engineering, Faculty of Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia
[3] CV Lapitaya, Yogyakarta, Indonesia
[4] Environmental Studies Center, Universitas Gadjah Mada, Yogyakarta, Indonesia
[5] Groundwater Working Group (GWWG), Universitas Gadjah Mada, Yogyakarta, Indonesia
[6] USAID IUWASH, Jakarta, Indonesia

[*]corresponding author: indraagusriyanto@unmaha.ac.id

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Abstract

Lake Toba is one of the prioritized conservation lakes in Indonesia, crucial for domestic needs, tourism, fisheries, agriculture, and power generation. However, Lake Toba's water levels and discharge have declined in recent decades. This study aimed to enhance water recharge and the water level of Lake Toba through hydrological modeling using the SWAT analysis and regional and specific conservation measures. The SWAT analysis utilized input data, including climate, soil, geomorphology, land use, and hydrology, to generate both existing and post-conservation water balance models. Conservation methods were categorized into civil engineering and vegetative approaches. Vegetative conservation techniques included agroforestry and MPTS (Multi-Purpose Tree Species), while civil engineering methods encompassed terracing, trenching, and infiltration wells. Regional conservation focused on land use plant types, while specific conservation involved detailed classification of the Lake Toba watershed by slope class, plant type, and land use. The analysis revealed significant land use changes in the Lake Toba catchment area, with dry fields increasing from 72,961 ha to 125,000 ha, a decrease in water level from 905 to 903 m above sea level, and a reduction in discharge from 180 m^3/s to 125 m^3/s. SWAT analysis shows that the existing inflow to Lake Toba is 131 m^3/s (with 78% accuracy), with potential increases to 250 m^3/s due to regional conservation and 180 m^3/s due to specific conservation. Conservation efforts have also improved discharge rates across 39 sub-watersheds of Lake Toba, both monthly and annually.

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Introduction

Lakes are one of the most widely used water sources in the world (Plisnier et al., 2024; Polidori et al., 2024), containing 0.3% of the earth's total fresh water supply (Musie et al., 2023). Lake water is commonly used for domestic purposes, livestock, agriculture, and power generation (Bakken et al., 2016; Lawrence et al., 2024). Several studies have shown that climate and land use change have impacted numerous lakes worldwide (Bergstrom et al., 2024; Mattheus et al., 2024), leading to lake shallowing and water level

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reduction. In addition, a considerable amount of lakes in the world are becoming polluted by agricultural and domestic waste (Perlov et al., 2023; Graham et al., 2024), causing ecosystem degradation and decreased lake water volume (Ivanova et al., 2024; Soreghan et al., 2024).

Indonesia is one of the countries with abundant lake freshwater sources, comprising 521 lakes throughout the country (Hidayat, 2019). Lakes in Indonesia are widely utilized to meet domestic water needs, agricultural demands, and electricity generation (Fadlillah and Widyastuti, 2016; Lukman et al., 2020; Widyastuti et al., 2024). However, lakes in Indonesia face significant challenges, such as water level decline, lake shallowing due to sedimentation, and pollution. Currently, 15 lakes in Indonesia are classified as critical, requiring immediate conservation and restoration efforts to preserve their sustainability (Indonesian Government Regulation, 2021).

One of the conservation priority lakes in Indonesia is Lake Toba, a volcanic-tectonic type lake (Lukman et al., 2019) that is currently experiencing significant challenges, including water level decline and pollution (Lukman et al., 2019; Hastuti et al., 2024). The water level decline in Lake Toba has occurred significantly due to massive land use changes within the Lake Toba watershed (Saputra et al., 2019), with the most significant transformation being forested areas that are being converted into dryland fields agricultural lands.

Acreman et al. (1993) explained that water balance studies indicate that Lake Toba's water level decline is mainly influenced by human activities, particularly land use changes, and only slightly influenced by climate factors (Irwandi et al., 2023). This finding aligns with a recent water balance study by Sihotang et al. (2012), who also stated that land use changes significantly impact Lake Toba's water level. From 1988 to 2020, substantial land use changes around Lake Toba included a 22% increase in settlement, a 9% increase in bare land, and a 29% decrease in forest areas (Rosid et al., 2023).

Land cover changes in Lake Toba's national strategic area from 2006 to 2018 also revealed a notable shift, with a percentage of 50.55% non-forest land use (plantations, rice fields, fields, and shrubs) that is larger than total forest land use of 49.45% (Sitindaon and Iskandar, 2020). These land use changes subsequently impacted the runoff coefficients of each area, thereby affecting the overall water balance of Lake Toba (Wesli, 2018). Irwandi et al. (2021) also noted that in addition to land use changes, human activities related to lake water utilization also contribute to the decline in Lake Toba's water level and discharge. Aziz et al. (2020) found that climate phenomena like ENSO and IOD do not increase rainfall around the Lake Toba area, thus having minimal impact on its water level changes. These studies indicated that Lake Toba's water level decline is primarily driven by human activities, specifically land use changes and water utilization, with climate factors playing a minor role.

Lake Toba is a vital water resource, supporting hydroelectric power generation in the area (Amien et al., 2019), a key tourism destination as part of the UNESCO Global Geopark (Anele and Sam-Otuonye, 2021), nearby local fishing industry, and providing water for the surrounding communities (Sibarani, 2022). Given its importance, further hydrological modeling and conservation studies are crucial for Lake Toba. Therefore, this study aimed to improve water recharge and increase the water level of Lake Toba. The hydrological modeling was conducted using the SWAT application, employing scenarios of existing conditions, regional conservation, and specific conservation measures.

Materials and Methods

Lake Toba is located at coordinates Zone 47 N 440000-520000 mE and 260000-320000 mN (Figure 1). Administratively, Lake Toba spans six regencies: Samosir, Toba, Karo, North Tapanulu, Simalungun, and Dairi. Lake Toba has an average annual rainfall of approximately 2,000-2,500 mm (Irwandi et al., 2023). Geologically, it consists of the Young Toba Tuff Formation, characterized by thick Ignimbrite rich in pumice (Chesner, 2012). The surrounding land use around Lake Toba includes dryland fields, plantations, and shrubland.

Soil Water Assessment Tools (SWAT) is one of the tools used for assessing the water balance within the watershed area (Francesconi et al., 2016). SWAT modeling is able to calculate flow components of water balance comprising evaporation, runoff, lateral flow, and base flow (Xie and Lian, 2013). It also allows changes in land use or conservation components in the model, which can alternate and calculate the water balance value obtained given the change. The data inputs for SWAT modeling include soil characteristics, climate, geomorphology, hydrology, and land use. The SWAT modeling process begins with data collection, followed by SWAT data parameterization, model simulation, and validation (Golmohammadi et al., 2017).

Geomorphological data were obtained from a 1:50,000 scale land system map, while the Digital Elevation Model (DEM) was derived from DEMNAS data from the Indonesian Geospatial Information Agency. Climate data, including rainfall (precipitation), relative humidity, temperature, wind speed, and solar radiation, was sourced from daily records by BMKG Climatology and Global Weather Stations for the years 2013-2022. Soil data was collected through field surveys, laboratory tests, and secondary data from the International Soil Reference and Information Center (ISRIC). The soil data used includes soil depth and texture, organic matter, bulk density, soil erodibility, soil permeability, coarse soil fraction, and soil albedo value (Riyanto et al., 2024).

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Hydrological data were obtained from secondary sources provided by PT Jasa Tirta and PT INALUM in the form of lake discharge and water level from the year 2003-2022. Additional hydrological data, such as water spring distribution and discharge was gathered through field surveys. Land use data was collected from both secondary data and field surveys, whereas secondary land use area data was acquired from the Ministry of Environment and Forestry (KLHK) for the years 1990, 2000, 2010, and 2020, while field surveys focused on assessing vegetation density and conservation method of the land use.

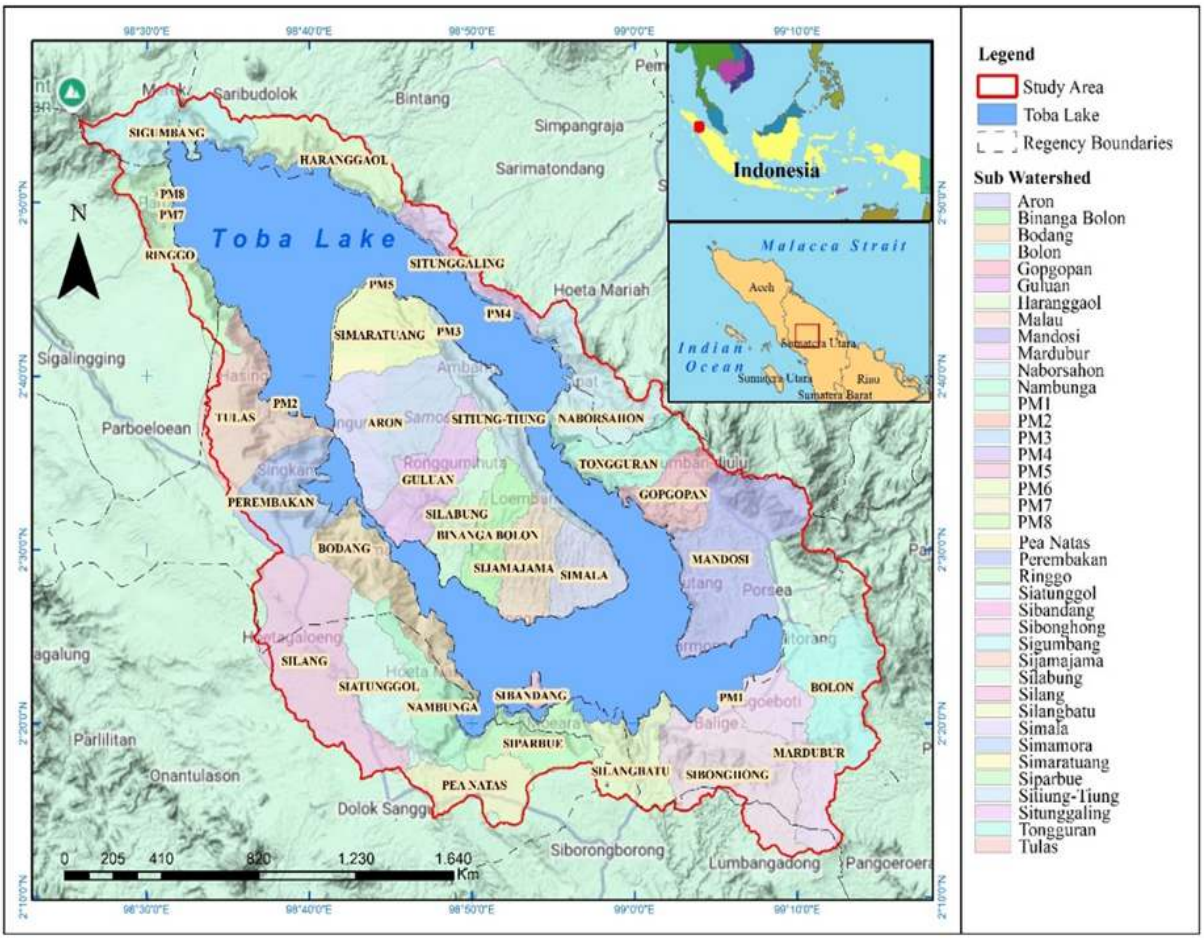

Figure 1. Research area.

The tools used in this study included ArcGIS Software for creating and displaying maps, SWAT Tools for water balance modeling, and Microsoft Excel for processing the SWAT modeling results. The materials used in this research comprised soil data, geomorphological data, climate data, land use data, hydrological data, and other data from previous studies, including water balance data, climate data, land use change, geology, and soil data. Surface runoff was calculated using the SCS-CN (Soil Conservation Service Curve Number) method, as described by Lai et al. (2020) in Equation 1.

Lateral flow was determined using the kinematic storage model developed by Nie et al. (2023) in Equation 2. Base flow in the model is defined as the flow contributing to the river volume in each sub-watershed, as demonstrated in Equation 3 (Jin et al., 2024). Model validation was performed by comparing model values with observational data values (Wu et al., 2020).

Qsurf = (Rday-Ia)^2/((Rday-Ia+S))……(1)

where :

Qsurf = runoff accumulation (mm)
Rday = rainfall depth (mm)
Ia = initial abstractions (mm)
S = retention parameter (mm)

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Qlat = 0.024 ((2.SWly, excess.Ksat.Slp)/(Ø d.Lhill))…..(2)

where :

Qlat = lateral flow discharge (mm H_2O/day)
SWly, excess = volume of water entering the saturated zone per unit area (mm H_2O)
Ksat = hydraulic conductivity (mm/hour)
Slp = slope steepness
Ø d = soil porosity (mm/mm)
Lhill = slope length (m)

Q_(gw) = ((8000.Ksat)/(L_gw^2)) x h_wtbl……..…(3)

where :

Q_(gw) = groundwater discharge into the river (mm H_2O/day)
Ksat = hydraulic conductivity (mm/day)
L_gw = slope length from the ridge to the river (m)
h_wtbl = groundwater table height (m)

The final outcome of SWAT analysis included surface runoff, lateral flow, and base flow for existing conditions, regional conservation, and specific conservation scenarios. Each flow value was provided as total discharge and discharge for each sub-watershed. Conservation activities were categorized into regional and specific types. Regional conservation was conducted based on slope classes and land use, incorporating both civil engineering and vegetative methods (Table 1). Regional conservation analysis was carried out by integrating maps of water resource vulnerability, forest and land status, critical land areas, and land use areas. The outcome of this analysis provided recommendations for regional conservation areas. Specific conservation focused on detailed assessments of spring sources through isotope testing and spring watersheds as outlined by Hendrayana et al. (2025). This type of conservation involved detailed methods regarding plant species and coverage percentages. The conservation scenario with the highest discharge value was used as a preferred alternative for conservation in the study area.

Table 1. Regional conservation.

Land Use	Vegetative Method	Civil Engineering Method
Shrubland	Agroforestry	
Highland Forest	Agroforestry	
Industrial Plantation Forest	Agroforestry	
Mixed Plantation	Agroforestry and MPTS plants	
Dryland Field	Agroforestry	Terracing and trenching
Bare Land		Terracing

Results and Discussion

The SWAT hydrological modeling mechanism was performed by characterizing the climate, slope parameters, soil, vegetation, and land use, and hydrology of the Toba Lake catchment area. Climate data indicated that rainfall in the Lake Toba area follows a monsoon pattern, with high rainfall occurring from January to April and October to December (Figure 2).

Maximum monthly rainfall reached approximately 475 mm, averaging 145 mm and totaling 1,500-2,500 mm annually, reflecting a high annual rate typical of tropical locations. In SWAT modeling, rainfall serves as a critical input for understanding the hydrological cycle and catchment response (Janjić and Tadić, 2023), as rainfall data directly affect outflow discharge and lake water levels. This rainfall data as an input component impacts Lake Toba's water discharge and water level as output components with higher rainfall typically correlated with increased water level and discharge rates, and vice versa (Figure 2).

Slope gradient plays a critical role in flow direction and accumulation within the hydrological cycle, influencing infiltration and runoff rates (Qi et al., 2020). Slope gradients in the Toba Lake catchment vary between 0 and >40% (Figure 3), with the majority falling in the 0-8% range, covering 143,906 ha, and the smallest area falling within the 25-40% range. The dominance of gentle slopes (0-15%) suggested significant infiltration potential in the Toba Lake catchment area, while the limited presence of steep and very steep slopes (25%) indicates a lower runoff potential. Therefore, it can be said that the slope gradient parameter indicates a higher infiltration than runoff potential in the Lake Toba catchment. This high infiltration potential identified areas that enhance groundwater recharge, which contributes to groundwater inflow increase to Lake Toba, while the low potential runoff identified areas that supported river inflow to Lake Toba.

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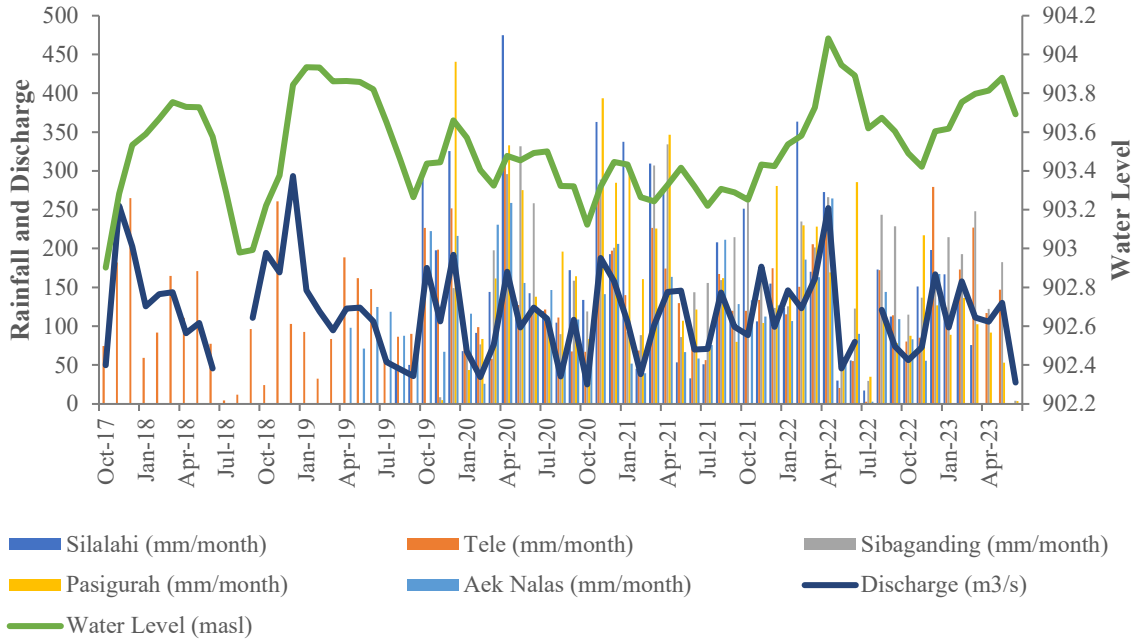

Figure 2. Hydrological components of Toba Lake.

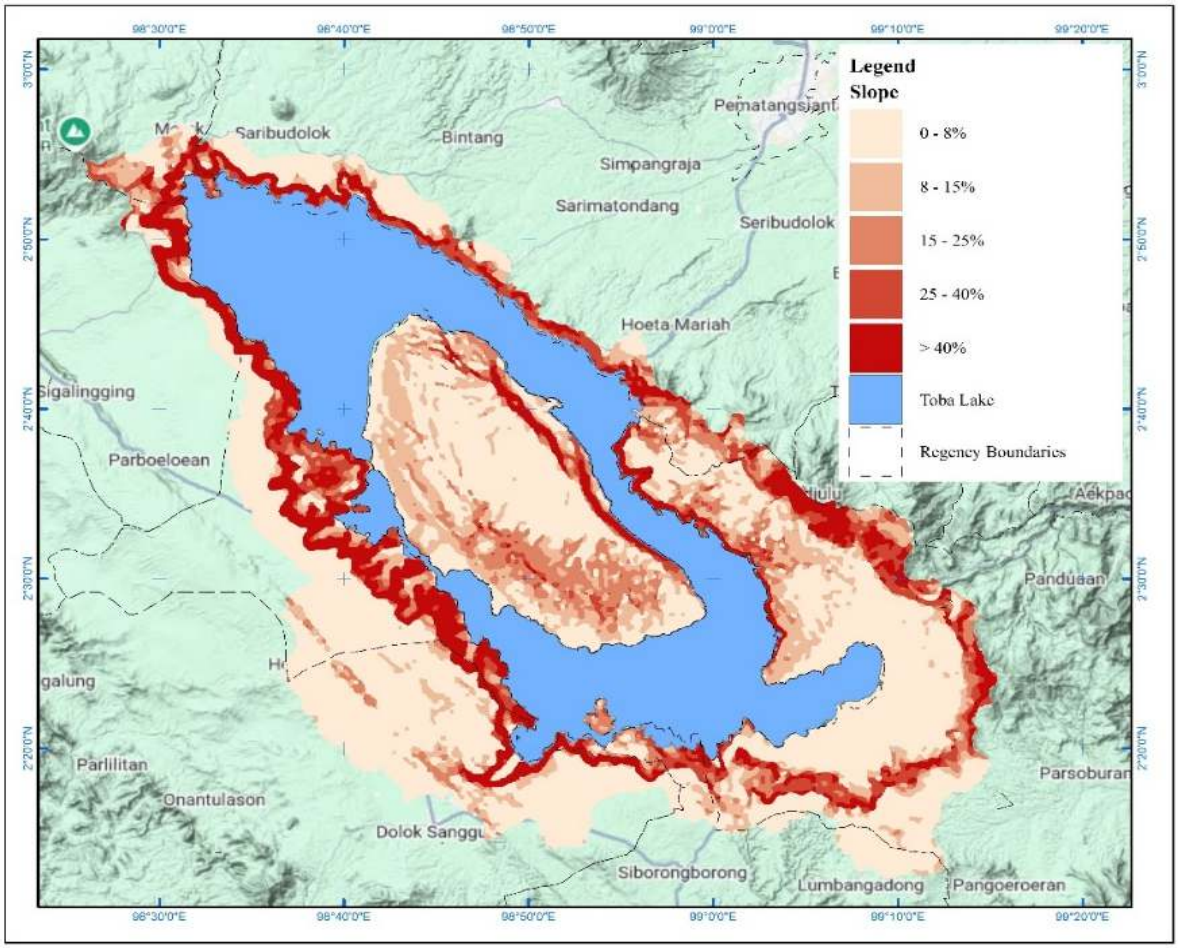

Figure 3. Slope of Toba Lake catchment.

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Worqlul et al. (2018) highlighted that soil physical characteristics significantly influence hydrological processes, including infiltration and runoff. The Toba Lake catchment exhibited considerable variation across several parameters, including soil organic material (SOM), permeability, texture, bulk density, and depth (Riyanto et al., 2024) (Figure 4). Despite the wide variations, analysis of 19 soil samples representing the Toba Lake Catchment indicated generally favorable conditions for infiltration and runoff. SOM values ranged from 0.66% to 12.89%, with the majority of it classified as high, demonstrating the soil's substantial capacity to retain water and enhance infiltration rates. Soil permeability classification also revealed high infiltration potential, with values ranging from 0.99 to 76.92 cm/hour, primarily in the moderate to very high categories. Additionally, the catchment's soil texture predominantly falls within the loam classification, known for its substantial water-holding capacity. Soil depth in the catchment varied between 55 and 200 cm, providing excellent capacity for water storage and transmission during infiltration and percolation in the unsaturated zone.

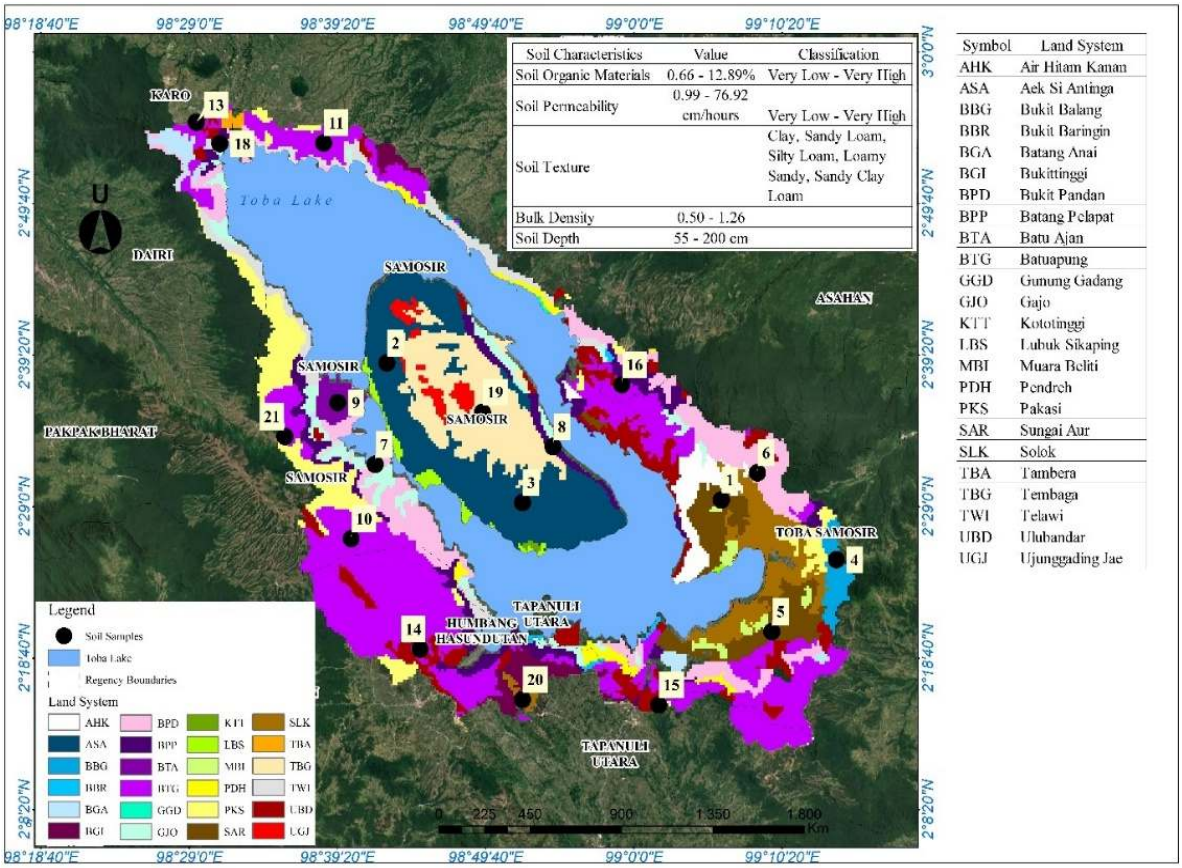

Figure 4. Soil characteristics of Toba Lake catchment (Riyanto et al., 2024).

In the Toba Lake catchment, the largest existing land use type is dry field, covering 125,404 ha (46.6% of total land use), while the smallest is bare land at 1,770 ha (0.65%) (Figure 5). The second largest land use is shrubland, encompassing 39,803 ha (14.79%), followed by forest at 33,968 ha (12.62%).

Overall, land use in the Toba Lake catchment is mainly composed of types that enhance infiltration, including forest, heterogeneous agriculture, paddy field, and dry field, while only a few types, such as bare land, settlement, and shrubland, contribute to increased runoff. Vegetation is believed to retain water through interception, throughflow, and stem flow (Sun et al., 2017), facilitating infiltration and minimizing runoff. In contrast, residential and bare land lacks vegetation and other factors to retain water, resulting in most water directly undergoing runoff to lower elevations. The decline in discharge and water level of Lake Toba is linked to increased rainfall and rising temperatures from 1993 to 2020 (Irwandi et al., 2023), shown by an upward trend of monthly rainfall data across all rain stations from 2013 to 2023. These climatic shifts were compounded by land use changes in the Lake Toba watershed from 1990-2020 (Figure 5), including a significant increase in dry fields from 72,961 ha in 1990 to 125,404 ha in 2020 (Figure 6), along with rises in bare lands, settlements, industrial plantation forests, and rice fields. These have affected the lake's water system and management, confirming the study by Chen et al. (2023), which suggested that updated land use data may improve the accuracy of SWAT model outcomes.

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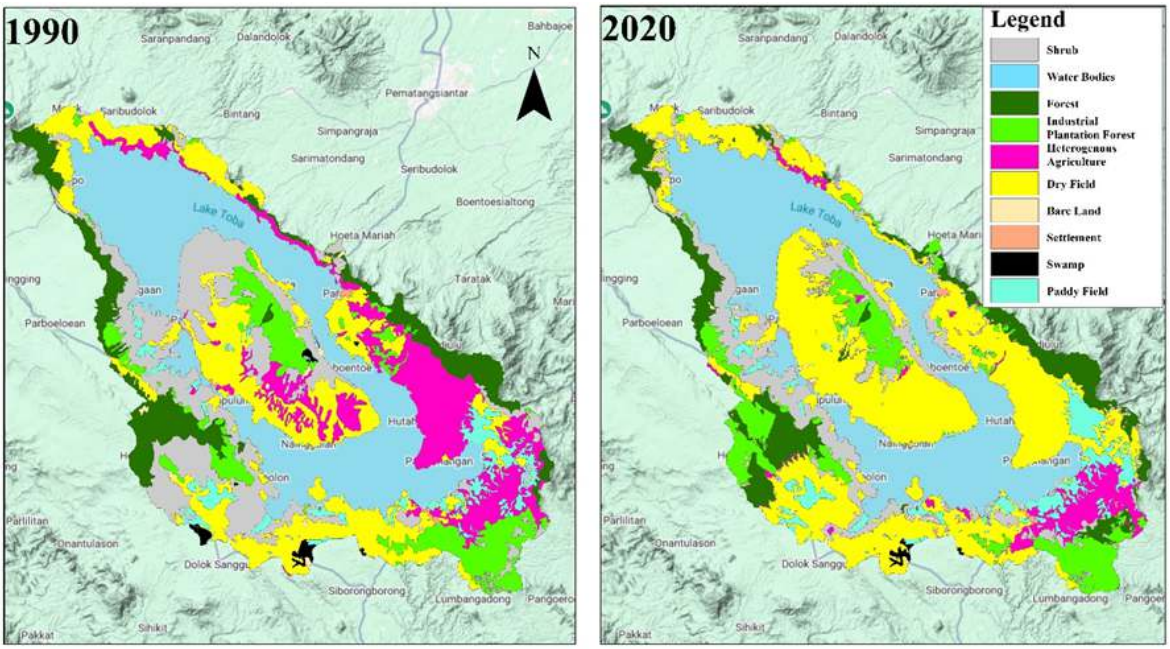

Figure 5. Land use change map of the Lake Toba watershed.

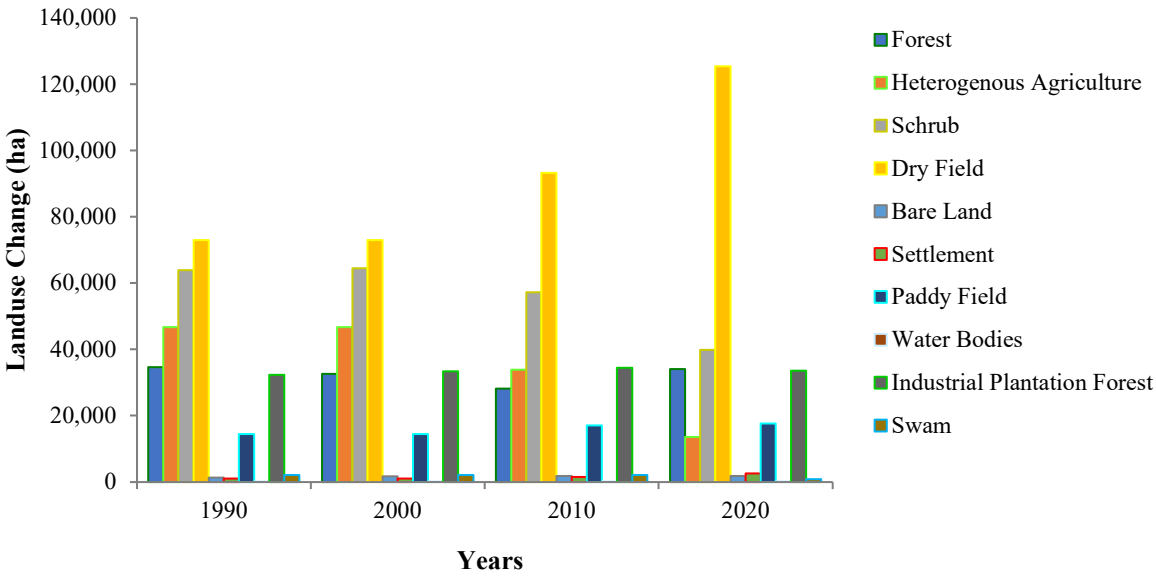

Figure 6. Land use change in Toba watershed (Sources: KLHK, 2020).

Automatic Water Level Recorder (AWLR) monitoring indicated that Lake Toba's water level has decreased from 905 meters above sea level (masl) between 2003-2015 to 903 masl between 2016-2013 (Figure 7), mirroring a similar pattern observed between 1960-2010 (Irwandi et al., 2021). Furthermore, the outflow discharge from Lake Toba to the Asahan River also decreased from 220-180 m^3/s during 2003-2016 to 105-125 m^3/s in 2017-2022. This consistent relationship highlighted how Lake Toba's water levels directly influence discharge rates. The geological conditions of the study location were found to be highly favorable for groundwater storage due to its water-bearing capacity, classified as an aquifer (Hendrayana et al., 2025). As an aquifer, these lithologies supply water to the lake rather than contributing to the decline in the lake water levels. These findings aligned with Abunada et al. (2021), who demonstrated that aquifers provide significant input to SWAT model results.

The area's soil characteristics are primarily composed of thick andosol, characterized by its high permeability, moderate organic content, and loamy texture (Riyanto et al., 2024), which allows effective and efficient water retention. These results were generally in line with Xiao et al. (2023), who found that certain soil types can enhance runoff in SWAT modeling.

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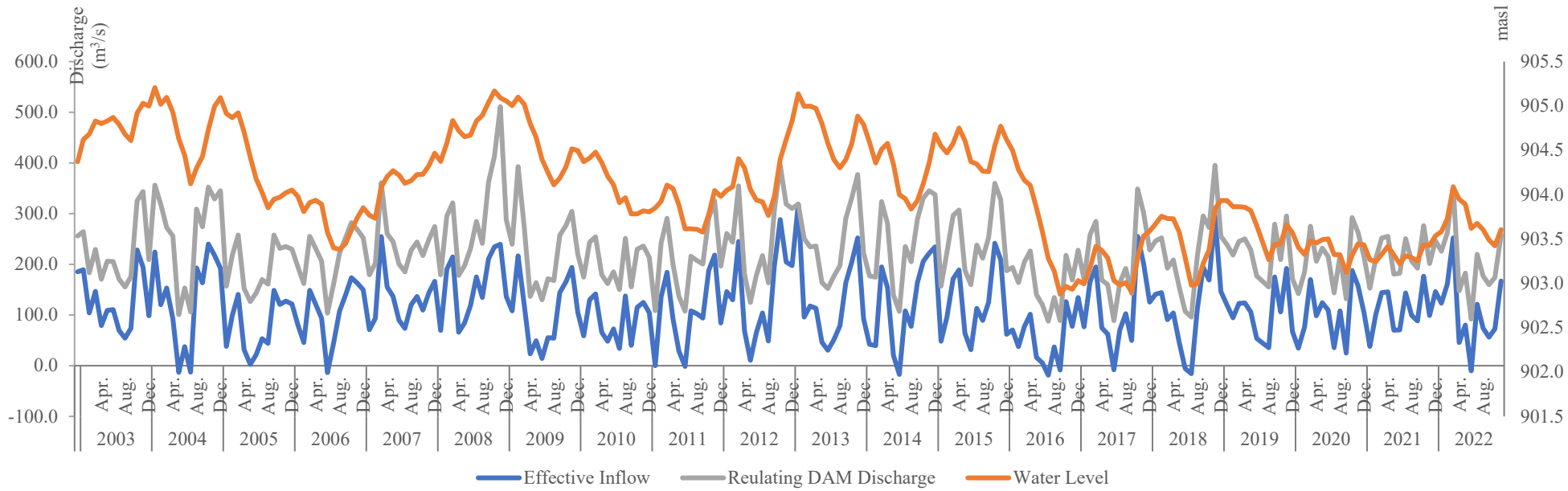

Figure 7. Discharge, water level, and DAM regulating graphic at Toba Lake (Sources: PT Jasa Tirta 1, 2023).

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The existing water balance was assessed using the SWAT model, which estimated an average inflow to the Lake Toba watershed of approximately 131.07 m^3/s (Table 2).

This monthly inflow estimation closely matched the discharge measurements of 107.03 m^3/s recorded by PT Inalum (Figure 8), achieving a validation accuracy of 78%. Chiang et al. (2023) and Jin et al. (2024) considered a validation accuracy range of 70-80% to be excellent in hydrological modeling, given the significant uncertainties often involved in hydrological modeling.

Table 2. SWAT model validation.

Description	SWAT Model Discharge	Measured Discharge
Monthly Average (m^3/s)	1,572.78	1,284.33
Daily Average (m^3/s)	131.07	107.03
Minimum Discharge (m^3/s)	68.37	30.31
Maximum Discharge (m^3/s)	210.63	198.05

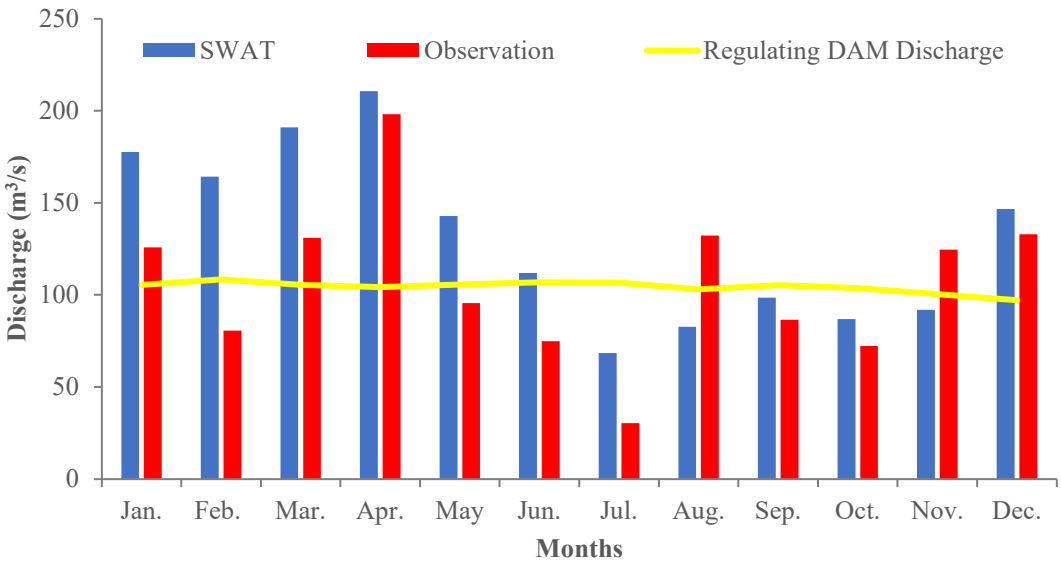

Figure 8. Comparison of SWAT model existing water balance with observation results.

Lake Toba watershed consisted of 39 sub-watersheds, each contributing varying inflows to the lake ranging from 1.70-8.31 m^3/s. The average surface runoff in these sub-watersheds varied between 0.48 and 3.35 m^3/s, with an average lateral flow between sub-watersheds of 0.34 m^3/s (Figure 9). The base flow in the model represented discharge from the shallow aquifer system, which contributes significantly to the main river flow discharge. The results of SWAT modeling estimated the groundwater potential of sub-watersheds in Lake Toba Watershed averages between 0.85-5.75 m^3/s. SWAT modeling also proposed that base flow is the largest contributor to discharge at 55.67 m^3/s, compared to surface runoff at 34.69 m^3/s and lateral flow at 10.53 m^3/s. The modeling results displayed that the Aron sub-watershed has the highest surface runoff, reaching 3.35 m^3/s. The primary factor contributing to high surface runoff in the Aron sub-watershed is the soil material characteristics, which are dominated by clay and dusty clay textures with deep to very deep soil depth classes, extending up to 104 cm. The average lateral flow between sub-watersheds in the Lake Toba watershed is 0.34 m^3/s, with the Ringgo sub-watershed having the highest lateral flow discharge of 0.81 m^3/s, while the Mardubur sub-watershed has the lowest lateral flow discharge of 0.14 m^3/s. Topographic characteristics and soil material types are the main factors influencing lateral flow formation (Abunada et al., 2021). The Ringgo sub-watershed features a hilly to mountainous topography with slopes ranging from moderately steep to very steep (25-60%), with soil characteristics dominated by clayey loam and sandy loam textures, enhancing its capacity to transmit substantial volumes of water.

Base flow in the SWAT model represented the discharge from the shallow aquifer system that contributed to the main river flow. SWAT modeling estimated that the average groundwater potential of the sub-watershed in the Lake Toba watershed ranges from 0.85-5.75 m^3/s. The Silang sub-watershed and Tulas sub-watershed exhibited the highest base flow potential, reaching 5.75 m^3/s and 3.01 m^3/s, while the Gopgopan sub-watershed has the lowest base flow potential at 0.85 m^3/s. The Silang sub-watershed has the largest discharge due to its predominantly gentle slopes (3-8%), clayey loam soil characteristics, a permeability of 0.72 mm/day, and plantation forests and secondary land forests as its main land use.

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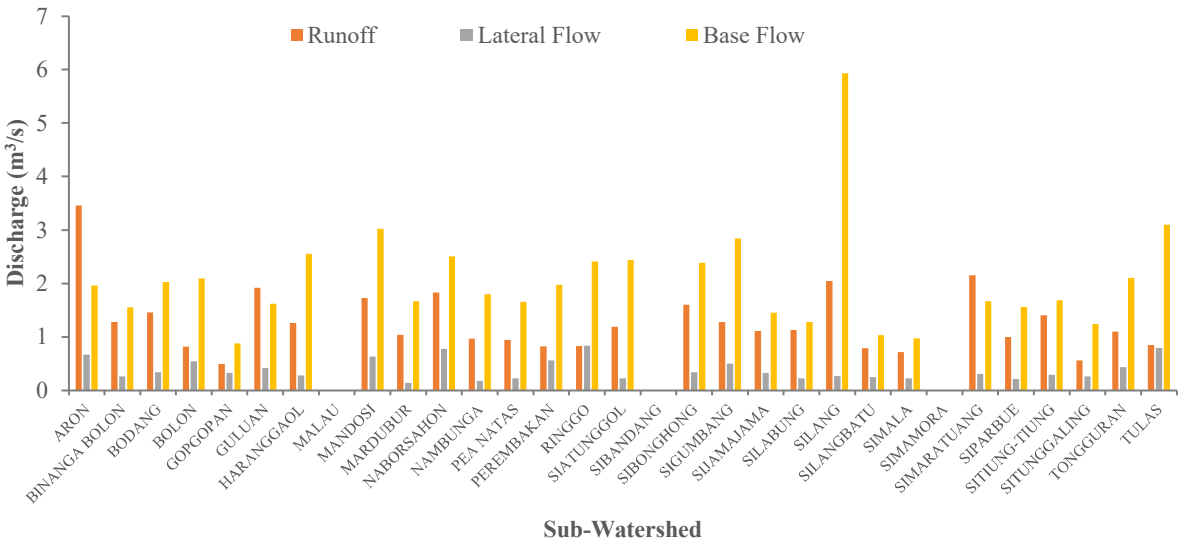
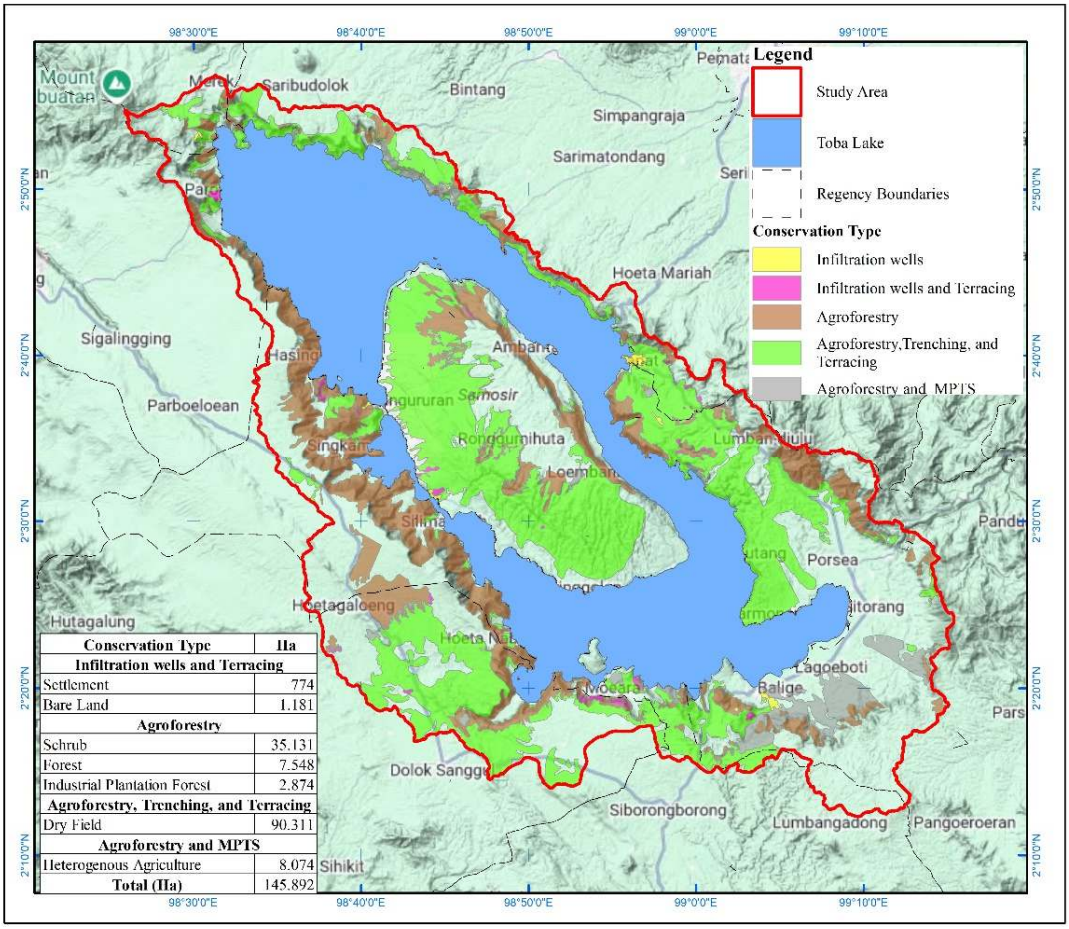


Figure 9. Existing water balance of Lake Toba.

Regional conservation scenarios used in SWAT modeling are categorized based on land use type. Two types of conservation methods were employed: vegetative methods and civil engineering methods. The land uses targeted for conservation included shrublands, bare lands, forests, plantations, settlements, and dryland fields. Vegetative conservation was implemented using agroforestry and MPTS (Multi-Purpose Tree Species) techniques, while civil engineering conservation methods included trenching, terracing, and infiltration wells that were applied in settlement areas (Figure 10).



Conservation Type	Ha
Infiltration wells and Terracing	
Settlement	774
Bare Land	1.181
Agroforestry	
Schrub	35.131
Forest	7.548
Industrial Plantation Forest	2.874
Agroforestry, Trenching, and Terracing	
Dry Field	90.311
Agroforestry and MPTS	
Heterogenous Agriculture	8.074
Total (Ha)	145.892

Figure 10. Lake Toba regional conservation.

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The implementation of conservation methods such as agroforestry, MPTS, and terracing on priority lands like dryland fields, bare lands, and shrublands have the potential to enhance flow components. This included an increase in base flow in the Lake Toba watershed, with groundwater storage rising by 10.56 m^3/s or 18% of the existing discharge (Table 3). Increased groundwater storage reduces surface runoff by 6.57 m^3/s or 18% of the existing discharge, and increases water yield from 105.68 m^3/s to 111.51 m^3/s. Regional conservation analysis results showed an increase of inflow entering the Lake Toba watershed by 143-387 m^3/s (Figure 11). The analysis also indicated that Lake Toba's discharge has shown a monthly increase under the implementation of regional conservation. The largest increase in discharge occurred in the Bolon sub-watershed at 79.66 m^3/s or 45% of the existing potential discharge. Additionally, discharge increases were also observed in the Mandosi, Mardbur, Penatas, Ringgo, Sibonghong, Silang, Silangbatu, Simala, Simarangtua, Situng-Tung, Tonggurai and Tulas sub-watersheds (Figure 12).

Specific conservation was detailly performed by categorizing conservation activities based on slope classes and land use into several codes, namely B1, B2, B3, H1, H2, H3, K1, K2, K3, L1, L2, L3, T1, T2, T3, P1, and P2 (Table 4). Each code has its own specific civil engineering and vegetative conservation method. The locations for specific conservation were aligned with the findings from watershed studies and isotope testing of springs as outlined by Hendrayana et al. (2025) (Figure 13).

The results of specific conservation analysis in Lake Toba indicated a potential increase in base flow from 57 m^3/s to 67 m^3/s and average water yield from 105.68 m^3/s to 113.65 m^3/s (Table 5). Specific conservation was able to effectively increase the average inflow to Lake Toba from 131.07 m^3/s to 180.41 m^3/s (Figure 14). This increase in lake inflow is attributed to the increased discharge from the main river in each sub-watershed. Notably, five sub-watersheds experienced significant increases in discharge, namely the Mandosi sub-watershed (51.18 m^3/s), Simaratuang (24.51 m^3/s), Ringgo (21.85 m^3/s), Pea Natas (18.31 m^3/s), and Nambunga (11.91 m^3/s) (Figure 15).

Overall, the results of both SWAT modeling based on regional and specific conservation demonstrated a potential increase in water inflow at Lake Toba. Similar patterns have been observed in other studies where conservation measures enhanced water balance results in SWAT modeling (Arabi et al., 2008; Melaku et al., 2018; Pandey et al., 2021). Comparable results have also been reported in other tropical regions of Indonesia, where conservation efforts improved water balance outcomes in SWAT modeling (Firdaus et al., 2014; Ridwansyah et al., 2014; Rachman et al., 2021).

Table 3. Changes in Lake Toba's watershed flow components after regional conservation.

Flow Components	Existing Condition (m^3/s)	Regional Conservation (m^3/s)
Average Runoff	35.79	29.22
Average Lateral Flow	10.86	10.55
Average Baseflow	57.44	68.00
Average Water Yield	105.68	111.51

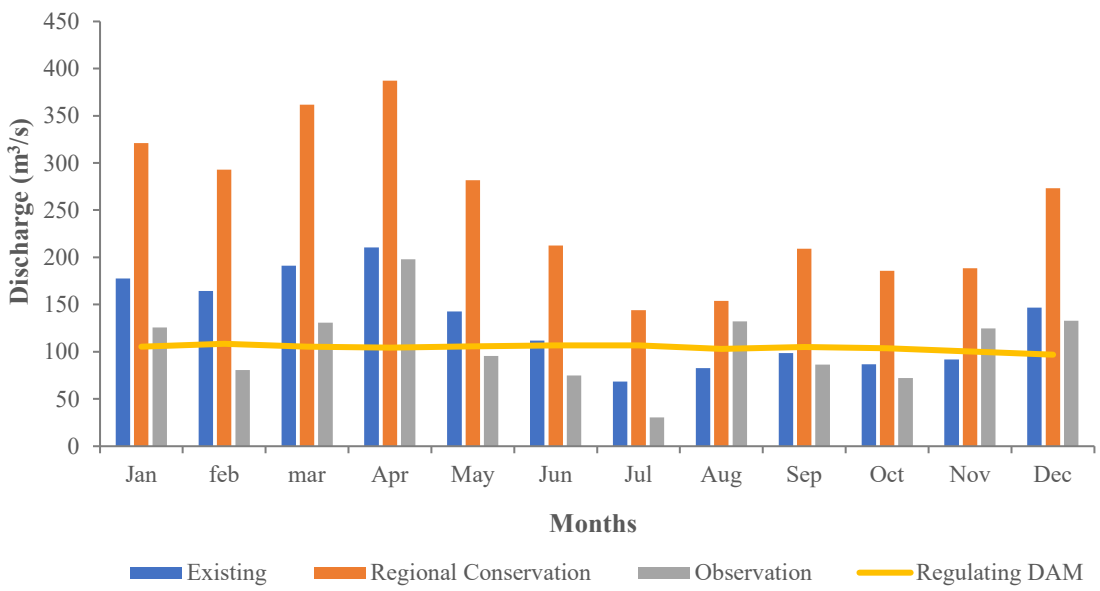

Figure 11. Comparison of Lake Toba watershed discharge between existing conditions and post-regional conservation conditions.

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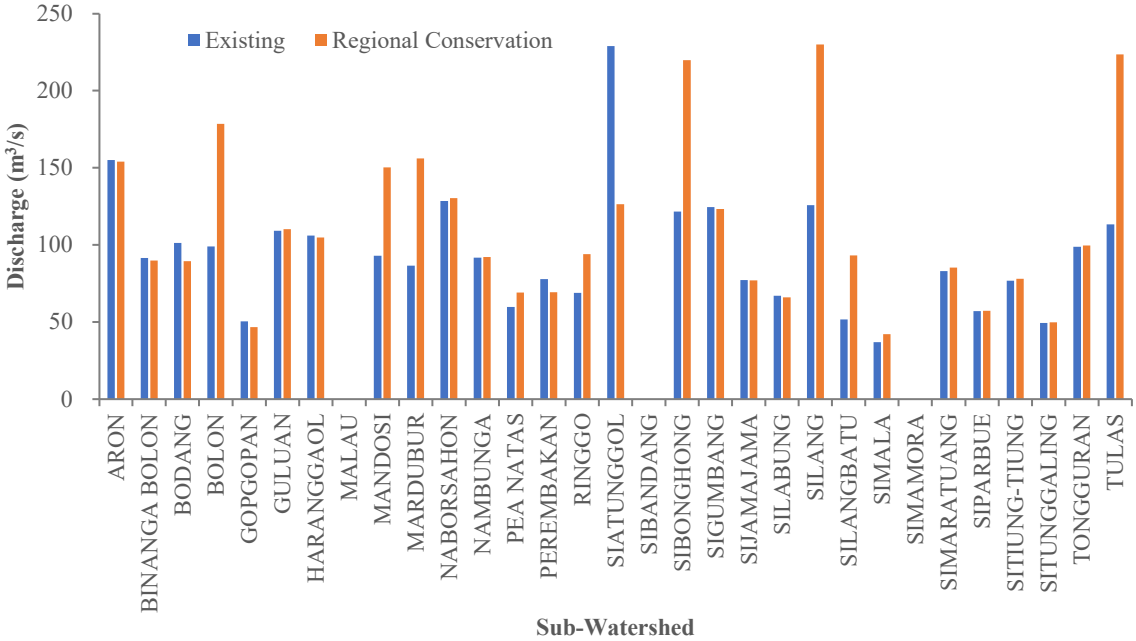

Figure 12. Comparison of Sub-watershed discharge in Lake Toba between existing conditions and post-regional conservation conditions.

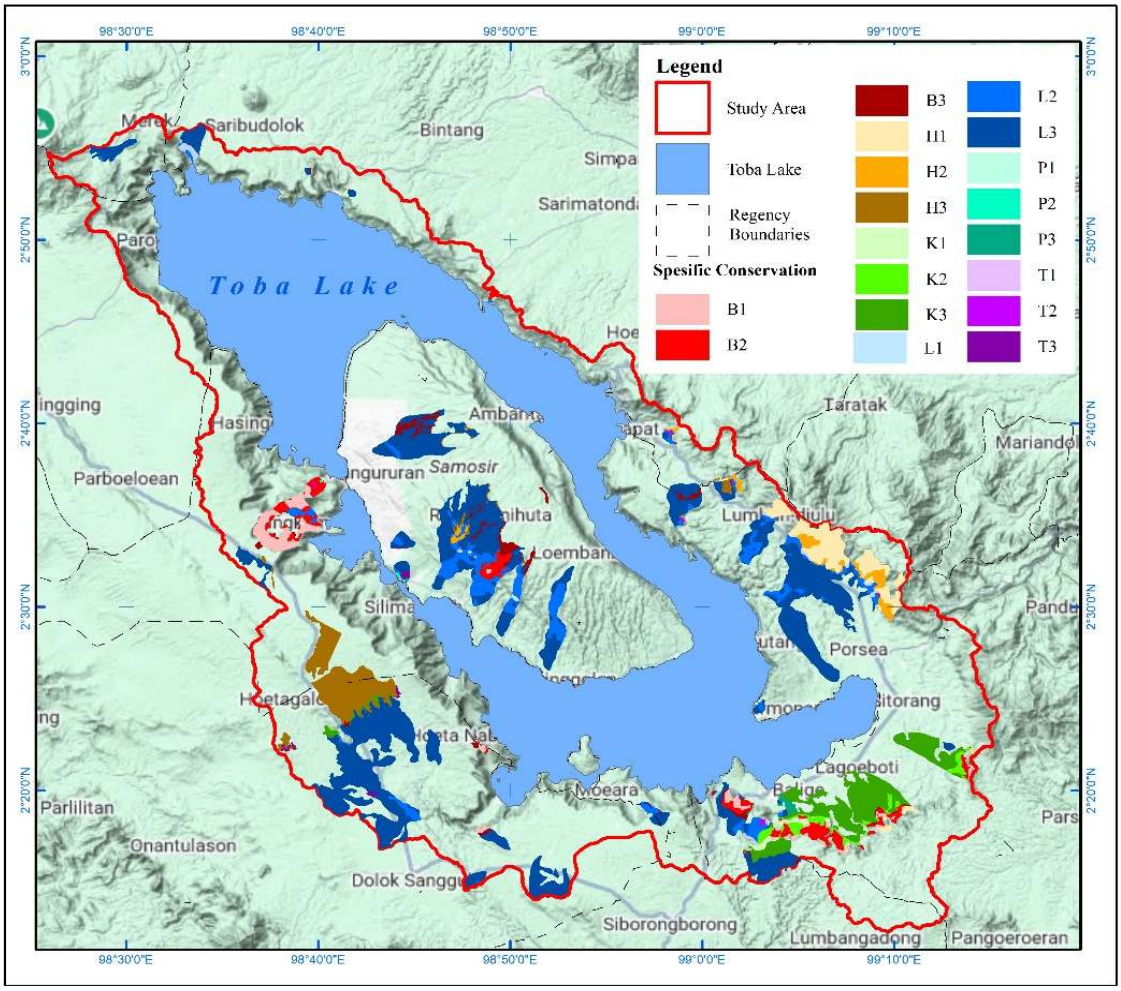

Figure 13. Lake Toba specific conservation.

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Table 4. Specific conservation methods: civil engineering methods and vegetative methods.

No	Landuse	Slope	Code	Civil Engineering Methods	Vegetative Methods
1	Shrubland	>40%	B1	Construction of drop structures and diversion channels along the surface water runoff	Reforestation of forest plants (*Gmelina*, *Shorea*, *Toona*) as canopy cover to cover up 60% of the area, with the remaining area designated for shrub crops (*Calliandra)* to stabilize forest plant slopes.
		15-40%	B2	Construction of terraces: level terrace, bench terrace, plantation terraces, and dead-end trenches. Construction of erosion control dams and structures along the surface water runoff	Agroforestry under the social forestry scheme with a ratio of 20% forest plants/MPTS fruit crops and 80% plantation crops (coffee. cacao) along with spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger) and grasses as cover crops.
		0-15%	B3	Construction of terraces: level terrace, bench terrace, plantation terraces, dead-end trenches, and biopore	Agroforestry under the social forestry scheme involves reforestation with a ratio of 20% forest trees and MPTS fruit plants as canopy cover, 30-35% plantation crops (coffee, cacao), and 40-50% seasonal food crops (vegetables, potatoes, peanuts, soybeans, etc.).
2	Forest and Industrial Plantation Forest	>40%	H1	Construction of drop structures and diversion channels along the surface water runoff	Reforestation of forest plants (*Gmelina*, *Shorea*, *Toona*) as canopy cover to cover up 20-60% of the area in accordance with forest function, with the remaining area allocated for planting shrub crops to stabilize forest plant slopes.
		15-40%	H2	Construction of erosion control dams and structures along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using forest plants (*Gmelina*, *Shorea*, *Toona*) as canopy cover to cover up 20-60% of the area, intercropping 20% of the area using plantation crops (coffee, cacao), with the remaining area allocated for planting shrub crops to stabilize forest plant slopes.
		0-15 %	H3	Construction of terraces: level terraces, bench terraces, plantation terraces, etc	Agroforestry under the social forestry scheme involves reforestation using forest plants (*Gmelina*, *Shorea*, *Toona*) as canopy cover to cover up 20-60% of the area, with the remaining area allocated for intercropping plantation crops (coffee, cacao).
3	Mixed Plantation	>40 %	K1	Construction of drop structures and diversion channels along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, while the remaining area is allocated for intercropping plantation crops (coffee, cocoa) and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger) as cover crops.
		15-40%	K2	Construction of terraces: level terrace, bench terrace, plantation terraces, and dead-end trenches. Construction of erosion control dams and structures along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, intercropping 60% area using plantation crops (coffee, cacao), with the remaining area allocated for with the remaining area allocated for spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).
		0-15%	K3	Construction of terraces: level terrace, bench terrace, contour terraces, dead-end trenches, and biopore	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, with the remaining area allocated for intercropping tuber and food crops (porang, taro, sweet potato, potato, peanuts, etc), horticultural crops (chili, tomato, cabbage, mustard), and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).

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No	Landuse	Slope	Code	Civil Engineering Methods	Vegetative Methods
4	Dryland Field	>40%	L1	Construction of drop structures and diversion channels along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, while the remaining area is allocated for intercropping plantation crops (coffee, cocoa) and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger) as cover crops to stabilize slopes.
		15-40%	L2	Construction of terraces: level terrace, bench terrace, plantation terraces, and dead-end trenches. Construction of erosion control dams and structures along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, intercropping 60% of the area using plantation crops (coffee, cacao), with the remaining area allocated for with the remaining area allocated for spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).
		0-15%	L3	Construction of terraces: level terrace, bench terrace, contour terraces, dead-end trenches, and biopore	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, with the remaining area allocated for intercropping tuber and food crops (porang, taro, sweet potato, potato, peanuts, etc), horticultural crops (chili, tomato, cabbage, mustard), and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).
5	Bare Land	>40%	T1	Construction of drop structures and diversion channels along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, while the remaining area is allocated for intercropping plantation crops (coffee, cocoa) and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger) as cover crops to stabilize slopes.
		15-40%	T2	Construction of terraces: level terrace, bench terrace, plantation terraces, and dead-end trenches. Construction of erosion control dams and structures along the surface water runoff	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, intercropping 60% of the area using plantation crops (coffee, cacao), with the remaining area allocated for with the remaining area allocated for spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).
		0-15%	T3	Construction of terraces: level terrace, bench terrace, contour terraces, dead-end trenches, biopore, infiltration wells	Agroforestry under the social forestry scheme involves reforestation using MPTS fruit plants (Avocado, Durian, Petai, Jackfruit) up to 20% of the area, with the remaining area allocated for intercropping tuber and food crops (porang, taro, sweet potato, potato, peanuts, etc), horticultural crops (chili, tomato, cabbage, mustard), and spices crops (turmeric, ginger, galangal, aromatic ginger, cardamom, torch ginger).
6	Settlement	>40%	P1	Stone terrace	None
		15-40%	P2	Stone terrace, biopore and infiltration wells	

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Table 5. Changes in Lake Toba's watershed flow components after specific conservation.

Flow Components	Existing Condition (m^3/s)	Specific Conservation (m^3/s)
Runoff	35.79	32.61
Lateral Flow	10.86	9.76
Baseflow	57.44	67.54
Water Yield	105.68	113.65

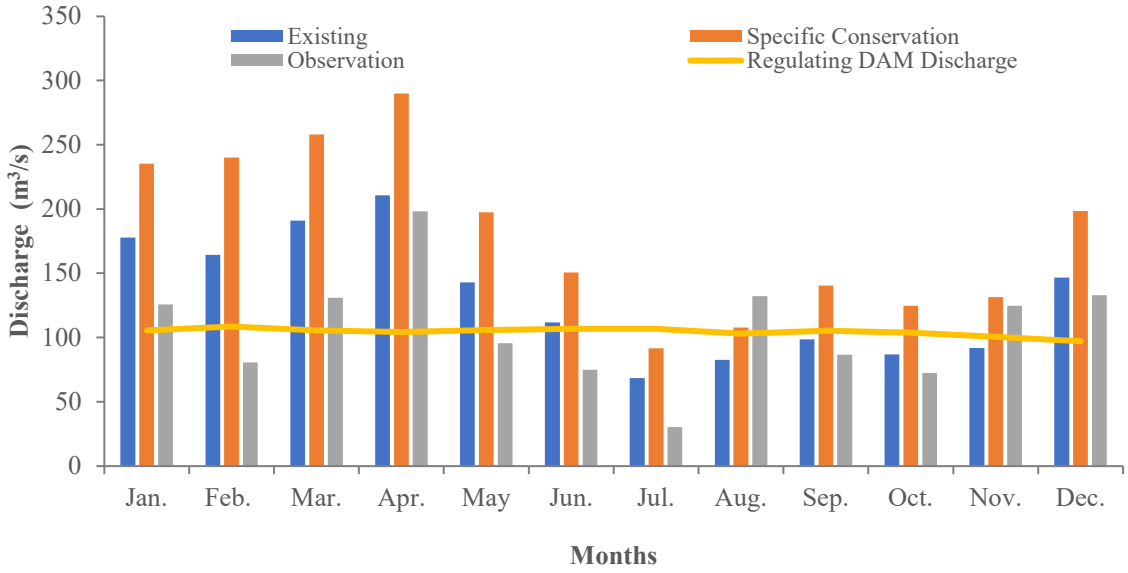

Figure 14. Comparison of Lake Toba watershed discharge between existing conditions and post-specific conservation conditions.

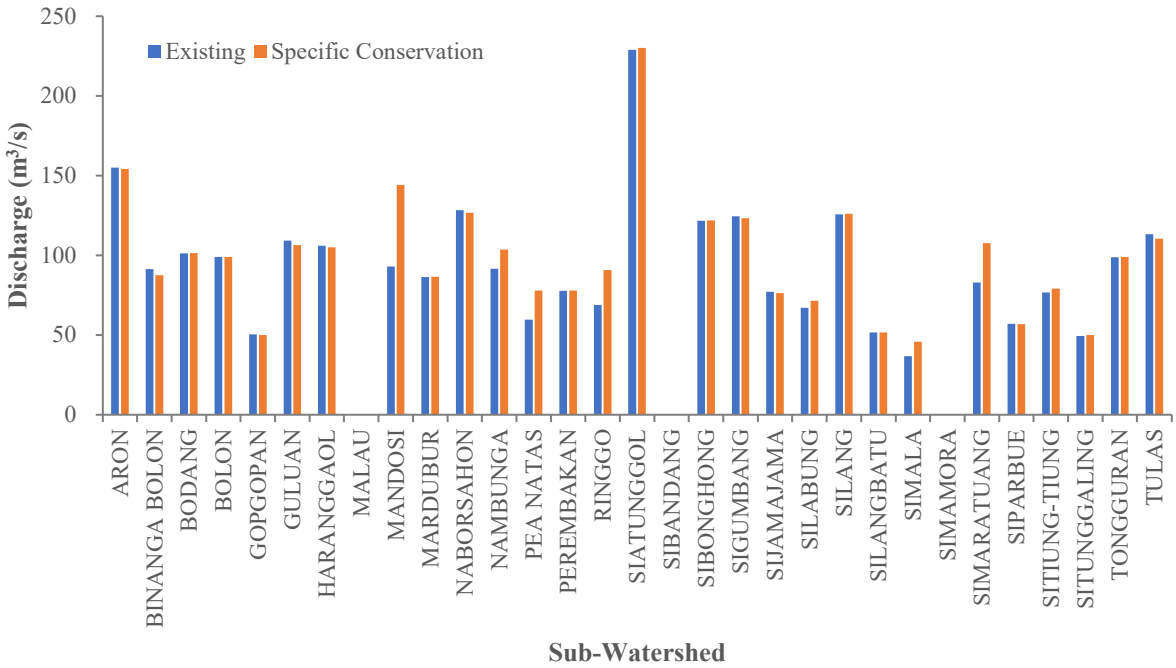

Figure 15. Comparison of sub-watershed discharge in Lake Toba between existing conditions and post-specific conservation conditions.

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Conclusion

The reduction in discharge and water levels of Lake Toba is attributed to land use changes and climate change, with hydrogeological, geological, and soil conditions identified as non-contributing factors. SWAT analysis indicated an existing inflow to Lake Toba of 131 m^3/s, which has the potential to increase to 250 m^3/s following regional conservation efforts and 180 m^3/s after specific conservation measures. Conservation activities recommended in a civil engineering manner include erosion retaining dams, terracing, trenching, infiltration wells, and drop structures, while vegetative conservation manners are recommended to utilize four types of plants: forest trees (woody plants), MPTS (in this case fruit plants), plantation plants, and shrubs.

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References

Abunada, Z., Kishawi, Y., Alslaibi, T.M., Kaheil, N. and Mittelstet, A. 2021. The application of SWAT-GIS tool to improve the recharge factor in the DRASTIC framework: Case study. *Journal of Hydrology* 592, doi:10.1016/j.jhydrol.2020.125613.

Acreman, M.C., Meigh, J.R. and Sene, K.J. 1993. Modelling the decline in water level of Lake Toba, Indonesia. *Advances in Water Resources* 16(4):207-222, doi:10.1016/0309-1708(93)90039-i.

Amien, S., Zelin, Y. and Fahmi, F. 2019. Relationship between water availability and power plant performance using water balance method: Case of PT. Inalum Asahan. *Simetrika Journal of Engineering and Technology* 1(1):22-29, doi:10.32734/jet.v1i1.670.

Anele, K.K, and Sam-Otuonye, C.C. 2021. Sustainable tourism: Evidence from Lake Toba in North Sumatra, Indonesia. *ASEAN Journal on Hospitality and Tourism* 19(01):52-62, doi:10.5614/ajht.2021.19.1.05.

Arabi, M., Frankenberger, J.R., Engel, B.A. and Arnold, J.G. 2008. Representation of agricultural conservation practices with SWAT. *Hydrological Processes* 22(16):3042-3055, doi:10.1002/hyp.6890.

Aziz, F., Kusratmoko, E. and Mandini, M.D. 2020. Estimation of changes in the lake water level and area using remote sensing techniques (Case study: Lake Toba, North Sumatra). *IOP Conference Series: Earth and Environmental Science* 561:012022, doi:10.1088/1755-1315/561/1/012022.

Bakken, T.H., Engeland, K., Killingtveit, A., Alfredsen, K. and Harby, A. 2013. Water consumption from hydropower plants: Review of published estimates and an assessment of the concept. *Hydrology and Earth System Sciences* 17(10):3983-4000, doi:10.5194/hess-17-3983-2013.

Bergstrom, R.D., Fergen, J., Johnson, L.B., Sterner, R.W., Lenters, J.D., Twiss, M.R., Steinman, A.D. 2024. Climate Change: A bibliometric study of the Great Lakes Basin. *Journal of Great Lakes Research* 50(3), doi:10.1016/j.jglr.2024.102316.

Chen, S., Fu, Y.H., Wu, Z., Hao, F., Hao, Z., Guo, Y., Geng, X., Li, X., Zhang, X., Tang, J., Singh, V. P. and Zhang, X. 2023. Informing the SWAT model with remote sensing detected vegetation phenology for improved modeling of ecohydrological processes. *Journal of Hydrology* 616, doi:1016/j.jhydrol.2022.128817.

Chesner, C.A. 2012. The Toba caldera complex. *Quaternary International* 258, doi:10.1016/j.quaint.2011.09.025.

Chiang, L.C., Shih, P.C., Lu, C.M. and Jhong, B.C. 2023. Strategies analysis for improving SWAT model accuracy and representativeness of calibrated parameters in sediment simulation for various land use and climate conditions. *Journal of Hydrology* 626, doi:10.1016/j.jhydrol.2023.130124.

Fadlillah, L.N. and Widyastuti, M. 2016. Water balance and irrigation water pumping of Lake Merdada for potato farming in Dieng Highland, Indonesia. *Environmental Monitoring and Assessment* 188(8), doi:10.1007/s10661-016-5452-7.

Firdaus, G., Haridjaja, O. and Tarigan, S.D. 2014. Hydrological response analysis for the application of soil conservation techniques in Lengkong sub-watershed using SWAT model. *Jurnal Ilmu Tanah dan Lingkungan* 16(1):16-23, doi:10.29244/jitl.16.1.16-23).

Francesconi, W., Srinivasan, R., Pérez-Miñana, E., Willcock, S.P., and Quintero, M. 2016. Using the Soil and Water Assessment Tool (SWAT) to model ecosystem services: A systematic review. *Journal of Hydrology* 535:625-636, doi:10.1016/ j.jhydrol.2016.01.034.

Golmohammadi, G., Rudra, R., Dickinson, T., Goel, P. and Veliz, M. 2017. Predicting the temporal variation of flow contributing areas using SWAT. *Journal of Hydrology* 547:375-386, doi:10.1016/j.jhydrol.2017.02.008.

Graham, M. and Ng, K. 2024. Concentrations and loads of metals, nutrients and organic contaminants entering the St. Lawrence River at Wolfe Island, 2000 to 2019. *Journal of Great Lakes Research* 50(3), doi:10.1016/j.jglr.2024.102340.

Hastuti, Y.P., Nirmala, K., Hutagaol, M.P., Tanjung, D., Kriswantriyono, A., Nurussalam, W., Wulandari, Y.P. and Fatma, Y.S. 2024. Analysis of main components of Lake Toba's water quality in different seasons. *Advances in Oceanography and Limnology* 15(1), doi:10.4081/aiol.2024.11726

Hendrayana, H., Riyanto, I.A., Nuha, A. and Satrio. 2025. Hydrogeology and hydrogeochemistry of Toba caldera aquifers, North Sumatra. *Indonesian Journal of Geography* 57(3):1-30.

Hidayat. 2019. Trend of rainfall over Indonesian major lakes from tropical rainfall measuring mission data. *IOP Conference Series: Earth and Environmental Science* 303(1), doi:10.1088/1755-1315/303/1/012019.

Indonesian Government Regulation. 2021. Number 60, National Lake priority. Deputy Secretary of the Cabinet for Legal Affairs and Legislation, Jakarta. https://peraturan.bpk.go.id/Details/171165/perpres-no-60-tahun-2021 (*in Indonesian*).

Irwandi, H., Rosid, M.S. and Mart, T. 2021. The effects of ENSO, climate change and human activities on the water level of Lake Toba, Indonesia: A critical literature

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review. *Geoscience Letters* 8(1):1-13, doi:10.1186/s40562-021-00191-x.

Irwandi, H., Rosid, M.S. and Mart, T. 2023. Effects of Climate change on temperature and precipitation in the Lake Toba region, Indonesia, based on ERA5-land data with quantile mapping bias correction. *Scientific Reports* 13(1), doi:10.1038/s41598-023-29592-y.

Ivanova, S.V., Fisk, A.T. and Johnson, T.B. 2024. Predicted growth of lake trout and Chinook salmon in a warming lake. *Journal of Great Lakes Research* 50(2), doi:10.1016/j.jglr.2024.102310.

Janjić, J. and Tadić, L. 2023. Fields of Application of SWAT Hydrological Model—A review. *Earth* 4(2):331-344, doi:10.3390/earth4020018.

Jin, L., Xue, H., Dong, G., Han, Y., Li, Z. and Lian, Y. 2024. Coupling the remote sensing data-enhanced SWAT model with the bidirectional long short-term memory model to improve daily streamflow simulations. *Journal of Hydrology* 634, doi:10.1016/j.jhydrol.2024.131117.

KLHK. 2020. Land use map shape file 1990, 2000, 2010, and 2020. The Ministry of Environment and Forestry, Jakarta. Dbgis.menklhk.go.id/arcgis/rest/services/KLHK.

Lai, G., Luo, J., Li, Q., Qiu, L., Pan, R., Zeng, X., Zhang, L. and Yi, F. 2020. Modification and validation of the SWAT model based on multi-plant growth mode, a case study of the Meijiang River Basin, China. *Journal of Hydrology* 585, doi:10.1016/j.jhydrol.2020.124778.

Lawrence, T.J., Achieng, A.O., Chavula, G., Haambiya, L.H., Iteba, J., Kayanda, R., Kaunda, E., Ajodendrayanan, M.Z., Muvundja, F.A., Nakiyende, H., Nyamweya, C., Obiero, K., Plisnier, P.D., Phiri, H., Sibomana, C. and Smith, S. 2023. Future success and ways forward for scientific approaches on the African Great Lakes. *Journal of Great Lakes Research* 49(6), doi:10.1016/j.jglr.2023.102242.

Lukman, L., Hidayat, H., Subehi, L., Dina, R., Mayasari, N., Melati, I., Sudriani, Y. and Ardianto, D. 2019. Pollution loads and its impact on Lake Toba. *IOP Conference Series: Earth and Environmental Science* 299(1), doi:10.1088/1755-1315/299/1/012051.

Lukman, L., Syawal, M.S. and Maghfiroh, M. 2020. Sumatran major lakes: Limnological overviews. *IOP Conference Series: Earth and Environmental Science* 535(1), doi:10.1088/1755-1315/535/1/012064.

Mattheus, C.R., Braun, K.N. and Theuerkauf, E.J. 2024. Morphodynamics and sand volumetric changes within engineered great lakes pocket beaches: Varied responses to a meter-scale lake-level rise, Chicago Lakefront. *Journal of Great Lakes Research* 50(2), doi:10.1016/j.jglr.2024.102302.

Melaku, N.D., Renschler, C.S., Holzmann, H., Strohmeier, S., Bayu, W., Zucca, C., Ziadat, F. and Klik, A. 2018. Prediction of soil and water conservation structure impacts on runoff and erosion processes using SWAT model in the northern Ethiopian highlands. *Journal of Soils and Sediments* 18(4):1743-1755, doi:10.1007/s11368-017-1901-3.

Musie, W. and Gonfa, G. 2023. Fresh water resource, scarcity, water salinity challenges and possible remedies: A review. *Heliyon* 9(8):2405-8440, doi:10.1016/j.heliyon.2023.e18685.

Nie, N., Li, T., Miao, Y., Zhang, W., Gao, H., He, H., Zhao, D. and Liu, M. 2023. Asymmetry of blue and green water changes in the Yangtze river basin, China, examined by multi-water-variable calibrated SWAT model. *Journal of Hydrology* 625, doi:10.1016/j.jhydrol.2023.130099.

Pandey, A., Bishal, K.C., Kalura, P., Chowdary, V.M., Jha, C.S. and Cerdà, A. 2021. A Soil Water Assessment Tool (SWAT) modeling approach to prioritize soil conservation management in river basin critical areas coupled with future climate scenario analysis. *Air, Soil and Water Research* 14, doi:10.1177/11786221211021395.

Perlov, D., Reavie, E.D. and Quinlan, R. 2023. Anthropogenic stressor impacts on hypolimnetic dissolved oxygen in Lake Erie: A chironomid-based paleolimnological assessment. *Journal of Great Lakes Research* 49(5):953-968, doi:10.1016/j.jglr.2023.04.006.

Plisnier, P.D., Kayanda, R., MacIntyre, S., Obiero, K., Okello, W., Vodacek, A., Cocquyt, C., Abegaz, H., Achieng, A., Akonkwa, B., Albrecht, C., Balagizi, C., Barasa, J., Abel Bashonga, R., … Lawrence, T. 2023. Need for harmonized long-term multi-lake monitoring of African Great Lakes. *Journal of Great Lakes Research* 49(6), doi:10.1016/j.jglr.2022.01.016.

Polidori, J.A., Paulson, H.L. and Gronewold, A.D. 2024. Assessing trends in urban municipal water use across the Great Lakes Basin. *Journal of Great Lakes Research* 50(1), doi:10.1016/j.jglr.2023.102243.

Qi, J., Lee, S., Zhang, X., Yang, Q., McCarty, G. W., and Moglen, G. E. 2020. Effects of surface runoff and infiltration partition methods on hydrological modeling: A comparison of four schemes in two watersheds in the Northeastern US. *Journal of Hydrology* 581, doi:10.1016/j.jhydrol.2019.124415.

Rachman, L.M., Nursari, E. and Baskoro, D.P.T. 2021. Application of SWAT in selecting soil and water conservation techniques for preparing management recommendations for Cilemer watershed, Banten, Indonesia. *IOP Conference Series: Earth and Environmental Science* 622:012023, doi:10.1088/1755-1315/622/1/012023.

Ridwansyah, I., Pawitan, H., Sinukaban, N. and Hidayat, Y. 2014. Watershed Modeling with ArcSWAT and SUFI2 in Cisadane catchment area: Calibration and validation to prediction of river flow. *International Journal of Science and Engineering* 6(2):12-21, doi:10.12777/ijse.6.2.92-101.

Riyanto, I.A., Hendrayana, H., Sambodo, A.P., Widyaningsih, Y. and Jayanto, G.D. 2024. Land suitability for coffee plants around Toba Lake. *Majalah Geografi Indonesia* 38(2):90-104, doi:10.22146/mgi.94042 (*in Indonesian*).

Rosid, M.S., Irwandi, H., Apip, and Mart, T. 2023. Investigation of main parameters causing decline in water level of Lake Toba during the last six decades. *Research Square*, doi:10.21203/rs.3.rs-3550465/v1/

Saputra, M.H. and Lee, H.S. 2019. Prediction of land use and land cover changes for North Sumatra, Indonesia, using an artificial-neural-network-based cellular automaton. *Sustainability (Switzerland)* 11(11), doi:10.3390/su11113024.

Sibarani, S. 2022. Waters utilization of toba lake for fish cultivation and tourism. *Proceedings of the International Conference on Culture Heritage, Education, Sustainable Tourism, and Innovation Technologies*, pp:558-564, doi:10.3390/su11113024.

Sihotang, H., Purwanto, M.Y., Widiatmaka, and Basuni, S. 2012. Model for water conservation of Toba Lake. *Jurnal Pengelolaan Sumberdaya Alam dan Lingkungan* 2(2):65-72, doi:10.19081/jpsl.2012.2.2.65 (*in Indonesian*).

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Sitindaon, M.H. and Iskandar, D.A. 2020. Land cover transformation in regional forest areas Lake Toba National Strategic Area. *Jurnal Planoearth* 5(2):88-94.

Soreghan, M., Cohen, A., McGlue, M., Yeager, K., Ryan, E., Johns, A. and Kimirei, I. 2024. Impacts of anthropogenic sedimentation on shell-bed habitats in Lake Tanganyika, Africa. *Journal of Great Lakes Research* 50(3), doi:10.1016/j.jglr.2024.102325.

Sun, L., Yang, L., Hao, L., Fang, D., Jin, K. and Huang, X. 2017. Hydrological effects of vegetation cover degradation and environmental implications in a semiarid temperate Steppe, China. *Sustainability* 9(2):1-20, doi:10.3390/su9020281.

Wesli, W. 2018. The decline normal water level of Lake Toba for integrated regional water management North Sumatera. *ARPN Journal of Engineering and Applied Sciences* 13(1):360-369.

Widyastuti, M., Haryono, E., Fadlillah, L., Afifudin, A., Cahyadi, A. and Agniy, R. 2024. Heavy metal risk assessment from agriculture in shallow tropical karst lake, Gunungsewu karst area, Java Island, Indonesia. *Iraqi Geological Journal* 57(1F):87-102, doi:10.46717/igj.57.1F.8ms-2024-6-17.

Worqlul, A.W., Ayana, E.K., Yen, H., Jeong, J., MacAlister, C., Taylor, R., Gerik, T.J. and Steenhuis, T.S. 2018. Evaluating hydrologic responses to soil characteristics using SWAT model in a paired-watersheds in the Upper Blue Nile Basin. *Catena* 163:332-341, doi:10.1016/j.catena.2017.12.040.

Wu, J., Yen, H., Arnold, J.G., Yang, Y.C.E., Cai, X., White, M.J., Santhi, C., Miao, C. and Srinivasan, R. 2020. Development of reservoir operation functions in SWAT+ for national environmental assessments. *Journal of Hydrology* 583, doi:10.1016/j.jhydrol.2020.124556.

Xiao, J., Wang, Y., Sun, J., Xie, S., Huang, Y., Wan, Z., Meng, L., Li, X. and Zhong, K. 2023. Effect of soil spatial aggregation caused by the calculation unit division on runoff and sediment load simulation in the SWAT model. *Journal of Hydrology* Volume 626, Part B, November 2023:130345, doi:10.1016/j.jhydrol.2023.130345.

Xie, H. and Lian, Y. 2013. Uncertainty-based evaluation and comparison of SWAT and HSPF applications to the Illinois River Basin. *Journal of Hydrology* 481:119-131, doi:10.1016/j.jhydrol.2012.12.027.

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